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(54) **The representation of character images in a compact form for computer storage.**

(57) There have been many approaches to the storage of representations of characters in a computer but no really satisfactory solution exists for storing the very large number of Chinese characters which are in use.

According to the invention the strokes of the characters are classified into regular strokes including horizontal, vertical and turn strokes mainly composed of straight segments, and irregular strokes in the form of curves, and different compressed representations are used respectively:

for regular strokes, a 24 to 48 bits compressed representation is used to describe the beginning X, Y coordinates, length, width, starting feature, ending feature turning feature and slant, and

for irregular strokes, e.g. of Chinese characters and characters of other languages, a series of vectors (approximately straight segments) are used to approximate to its curved contour with each vector represented by directed increments of X, Y coordinates and a number of different cases of increments are discriminated to make the representation more compact.

Such an arrangement provides compact storage particularly since it is possible to scale the stored representation to provide from a single basic size any other desired size.

The representation can be transformed into a dot matrix which is in turn used to form a page layout to provide input for a laser typesetting system.

EP 0 095 536 A1

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This invention relates to the compressed representation of high resolution characters, especially Chinese characters and other ideographic characters, in computer storage and to the transforming of this compressed representation into a high resolution dot matrix in a manner especially suitable for raster output.

This invention is particularly useful in connection with a fast, high resolution, character generator and typesetting controller in a computerized laser system for typesetting books and newspapers in Chinese as well as in other languages, but it also has application to a CRT typesetter.

Up to now, it has been very difficult to represent efficiently high resolution Chinese character images in computer storage. In book and newspaper publications, high printing quality is of prime importance and as a result resolution should be over 25 lines/mm. Moreover, numerous typefaces and sizes are needed in typesetting. The most advanced fourth generation laser typesetter has a series of advantages including the prospect of direct plate-making, but there are some problems caused by the raster output nature of laser system. Unlike a CRT, a laser output device cannot change its dot size instantly during scanning and therefore different sizes of the same character must be represented by different dot matrices. If the resolution is 29.3 lines/mm, various sizes of characters needed in typesetting Chinese newspapers and their corresponding dot matrices are as follows:

character size		dot matrix
in point	in mm^2	
56	19.66	576 x 576
48	16.86	494 x 494
42	14.74	432 x 432
36	12.63	370 x 370
31.5	11.06	324 x 324
28	9.83	288 x 288
21	7.37	216 x 216
18	6.28	184 x 184
15.75	5.53	162 x 162
14	4.91	144 x 144
12	4.23	124 x 124
10.5	3.68	108 x 108
9	3.14	92 x 92
7.875	2.76	82 x 82
7	2.46	72 x 72
6	2.12	62 x 62

Over ten typefaces are frequently used in Chinese publications. They are Song, imitation Song, display Song (for headline), boldface, regular script, long Song, short Song, long boldface, short boldface, long imitation Song etc. Each typeface must contain over 8,000 different Chinese characters. As a result the total storage requirement for all the Chinese characters in different typefaces and sizes used in Chinese publications would be well over 2×10^9 bytes. Therefore efficient representation of high resolution Chinese characters becomes a crucial problem in developing a Chinese typesetting system.